Mustafa Arda Aydın

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About Me

I am an M.S. student at Bilkent University specializing in medical image analysis. I focus on the efficient adaptation of vision-language and foundation models across diverse clinical domains, with a particular interest in model interpretability and decision-making mechanisms.

Education

Bilkent University, Ankara, Turkey 09/2024 - Present
Master of Science in Electrical and Electronics Engineering ***GPA: 3.91/4.00***

Istanbul Technical University, Istanbul, Turkey 09/2019 - 06/2024
Bachelor of Science in Electronics and Communication Engineering ***GPA: 3.94/4.00 (Rank:2/134)***

AGH University of Science and Technology, Kraków, Poland 02/2022 - 07/2022
Erasmus+ Student in Electronics and Communication Engineering

Work Experience

Imaging and Computational Neuroscience (ICON) Lab, Ankara, Turkey 09/2024 - Present

Graduate Research Assistant

- **Vision-Language Models (VLMs):** Developed an efficient, spatially aware adaptation framework for medical VLMs under Prof. Tolga Çukur. Consistently outperformed baseline fine-tuning techniques across multiple downstream tasks, including zero-shot classification, object detection, and semantic segmentation [*Accepted at NeurIPS Med-Reasoner Workshop 2026*].

ITU Vision Lab, Istanbul, Turkey 08/2022 - 09/2024

Undergraduate Researcher

- **Training-free Open-Vocabulary Semantic Segmentation:** Developed a method to transform image-level knowledge in vision-language models (VLMs) into pixel-level representations through image-, text-, and architecture-level enhancements. Achieved state-of-the-art performance on multiple benchmarks for training-free semantic segmentation, resulting in a publication at CVPR Workshops 2025.
- **Lesion Segmentation:** Participated in the BONBID-HIE MICCAI 2023 challenge. Developed a novel transformation method that significantly improves the dice score for this lesion segmentation task and successfully presented our challenge solution at the BONBID-HIE2023 MICCAI Workshop.
- **Tiny Object Segmentation:** Developed a successful image segmentation model and compared some of the state-of-the-art image segmentation models for the bird's-eye view of aerial images of the campus under the supervision of Prof. Gözde Ünal.

ASELSAN, Ankara, Turkey 07/2023 - 08/2023

Summer Intern at Autonomous Vehicle Team

- **Semantic Segmentation:** Worked on the possible implementation of the Segment Anything Model (SAM) for autonomous vehicles efficiently. Reduced the inference time and increased the quality of masks for some real-world applications.

Selected Publications

1. M. Arda Aydın, Melih B. Yılmaz, Aykut Koç, and Tolga Çukur, "ACE-LoRA: Graph-attentive context enhancement for parameter-efficient adaptation of medical vision-language models," in *NeurIPS Workshop on Medical Reasoning with Vision Language Foundation Models*, 2026. (*arXiv:2603.17079*).
2. M. Arda Aydın, Efe M. Cirpar, Elvin Abdinli, Gozde Unal, and Yusuf H. Sahin, "ITACLIP: Boosting training-free semantic segmentation with image, text, and architectural enhancements," *in Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR) Workshops*, pp. 4142–4152, 2025
3. M. Arda Aydın, Elvin Abdinli, and Gozde Unal, "Segresnet based reciprocal transformation for bonbid-hie lesion segmentation" in *Trauma Thompson Challenge*, pp. 39–44, Springer, 2023.

Presentations

1. 1st BOston Neonatal Brain Injury Dataset for Hypoxic Ischemic Encephalopathy (BONBID-HIE) Lesion Segmentation Workshop (2023)
2. Computer Vision and Pattern Recognition Conference Workshops 2025, Nashville, TN, USA
3. NeurIPS Workshop on Medical Reasoning with Vision Language Foundation Models 2026, Atlanta, GA, USA

Teaching Experience

Bilkent University, Ankara, Turkey 09/2024 - Present

Teaching Assistant

- **EEE 211 Analog Electronics:** Asking interview questions to students during laboratory sessions and grading the experiment reports.
- **EEE 102 Digital Logic Design:** Conducted student interviews during laboratory sessions, provided guidance on course projects, and graded laboratory reports.

Honors & Awards

- **Travel Grant:** Awarded a travel grant by the Scientific and Technological Research Council of Turkey (TÜBİTAK) to attend CVPR 2025.
- **Bilkent University Departmental Fellowship:** Full tuition waiver and stipend during the MSc program.
- **Best Paper Award Runner-Up:** Image Processing and Computer Vision Track, 34th Signal Processing and Communications Applications Conference (SIU 2026).
- **ITU Electronics and Communication Engineering Department:** Ranked second in the electronics and communication engineering department.
- **Erasmus+ Grant:** Granted to cover basic expenses during the exchange program (€1800)
- **Nationwide University Entrance Exam:** Ranked in the 0.1 percent among 2.5 million students in Turkey (2019)

Related Coursework

- Machine Learning
- Neural Networks
- Probability and Statistics
- Computer Vision
- Linear Algebra and Applications
- Medical Imaging
- Deep Generative Networks
- Deep Learning

Skills

- **Technical Skills:** Python, C, C++, MATLAB, PyTorch, TensorFlow, Deep Learning, Computer Vision
- **Languages:** Turkish (Native), English (Full Proficiency)